Is the L=3 knee serial composition, or parameters?

N=24 (d=5, q=4), 150 epochs, n_train=20000, d_model=256 -- only (num_layers, dim_feedforward) varies

49510197 trained, 49521537 scored on the shared pool at n=1500

SERIAL. At equal parameters, depth wins by a mile.

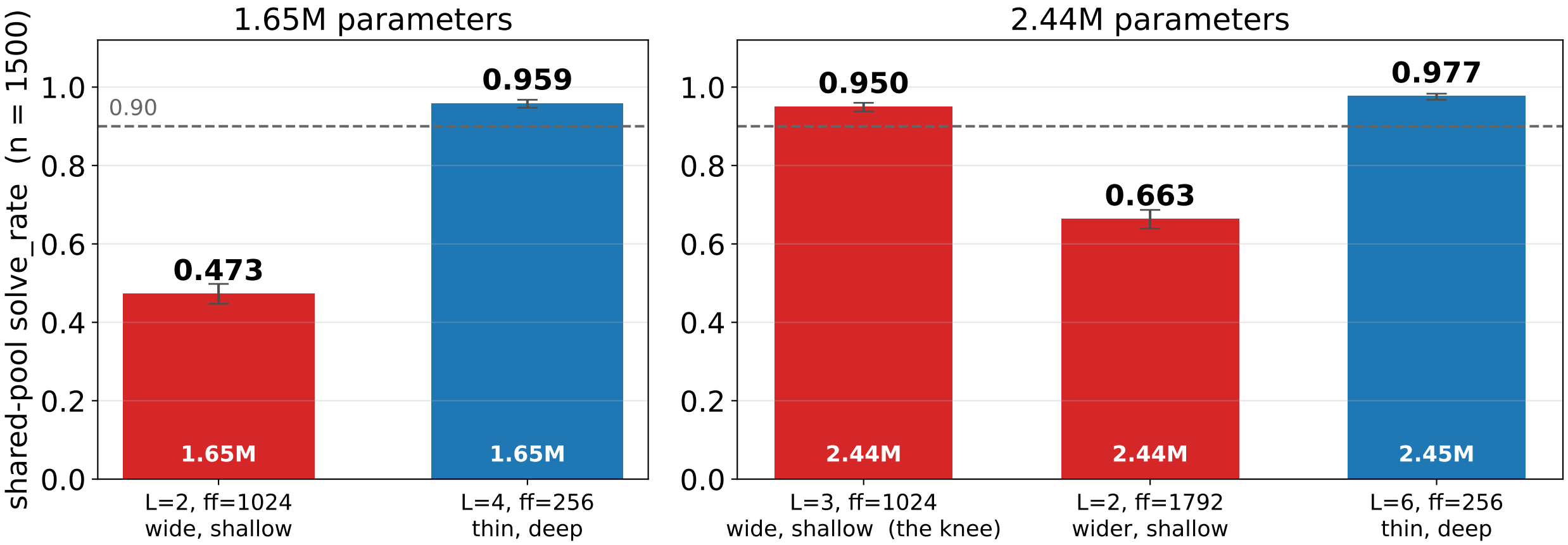
At 1.65M params, L=4/ff=256 scores 0.959 against L=2/ff=1024's 0.473
-- a gap of +0.486, paired p = 1e-203.

At 2.44M, depth still wins: 0.977 vs 0.950 (+0.027, p = 3e-05),
and no MLP width tested buys the knee at L=2 at any budget.

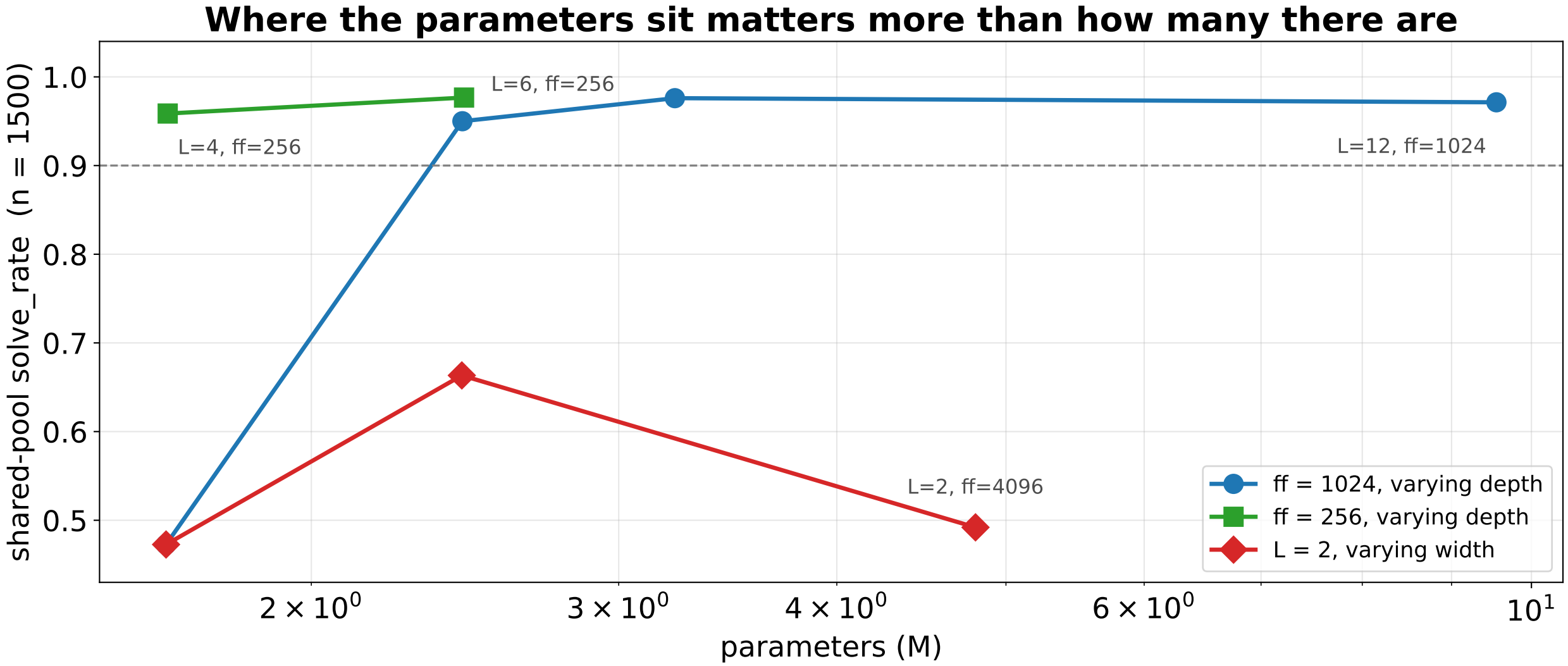
L=2/ff=4096 carries 4.8M -- 1.97x the knee's budget -- and scores 0.492.

Finding 11 established that L*(T) is flat at 3 and that L=2 already fails at T=1, so the third layer is a fixed cost paid before any chaining. That left two readings: H1, the cost is CAPACITY and where the parameters sit does not matter, so a wider MLP at L=2 should recover it; H2, the cost is a SEQUENTIAL step and no width buys it. The two make opposite predictions at matched parameters, which is what this deck measures. H1 is dead.

The same budget, spent two ways

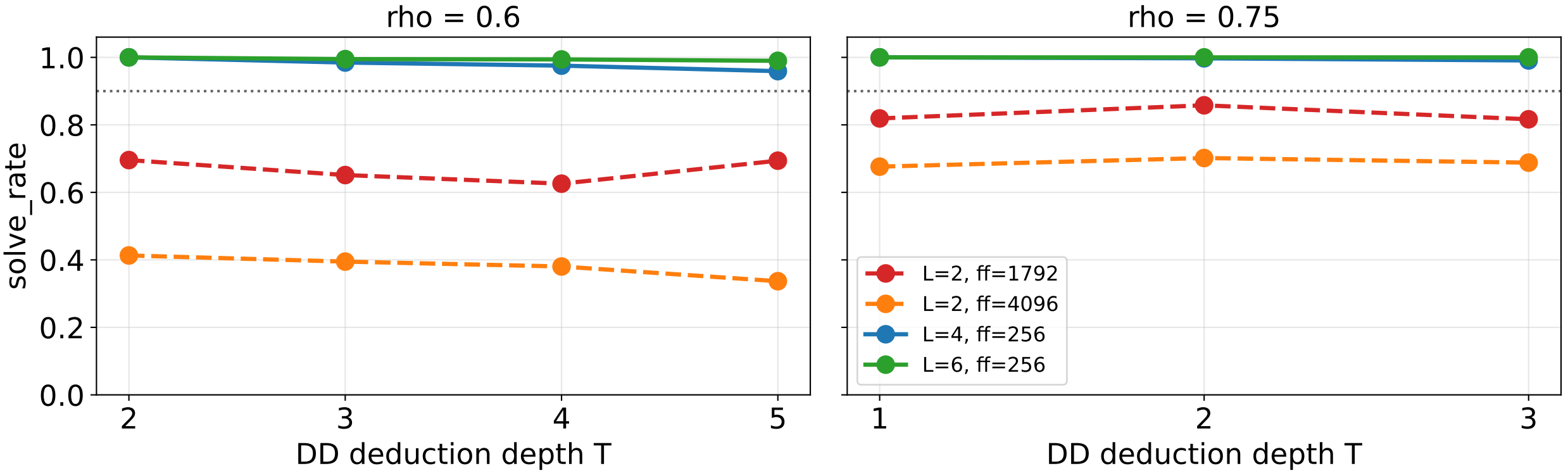


Each panel holds the parameter count fixed to 0.22% and moves it between depth and MLP width. Left is the discriminator: same budget, and the thin-deep cell clears 0.90 while the wide-shallow one does not clear 0.50 (+0.486, 739 discordant instances won to 10 lost, exact p = 1e-203). Under H1 these two should have landed within 0.10 of each other. Right repeats it at the knee's own budget, where the wide-shallow cell is the measured knee rather than a cripple, and depth still wins (+0.027, 65-25, p = 3e-05). Spending the knee's budget on width instead (L=2/ff=1792) gives 0.663: better than L=2/ff=1024, nowhere near the knee. Wilson intervals at n=1500; the paired McNemar is the test, since all cells are scored on the same instances.



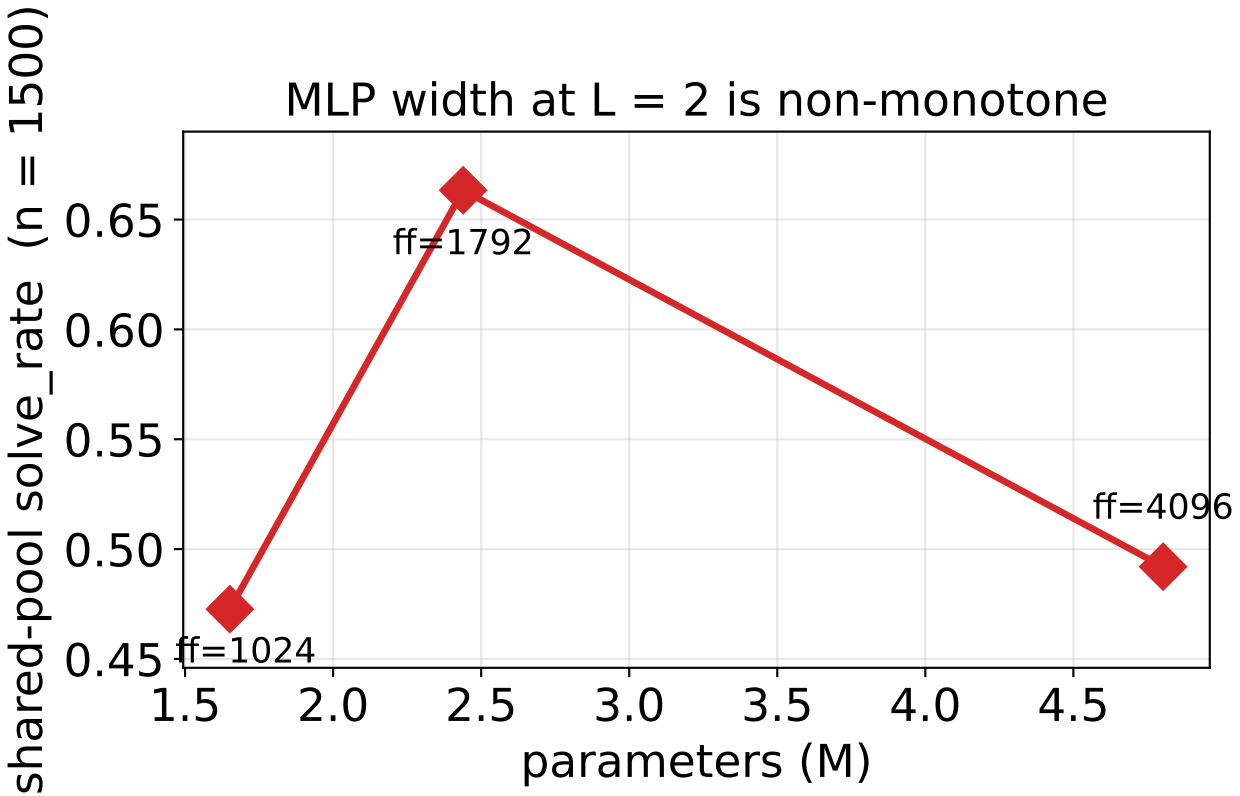
Three ladders through the same parameter axis. Only the two that add LAYERS climb to the ceiling; the one that adds WIDTH at L=2 rises to 0.663 and then falls back, never approaching 0.90, despite reaching 4.8M. The ff=1024 ladder starts at L=2 here because its L=1 cell (0.067) was not worth a shared-pool pass. The green ladder is the efficient one: L=6/ff=256 at 2.45M scores 0.9767, statistically tied with L=12/ff=1024 at 9.55M (+0.0053, 34 to 26, p = 0.37) and with L=4/ff=1024 at 3.23M (+0.0007, p = 1.00) -- the top of the reference ladder reached with 3.9x fewer parameters. That tie is the pre-registered (c) and it held. Caveat on reading this as a scaling law: ff=256 was tested at two depths only, so the green ladder is two points, and nothing here says ff can be cut further.

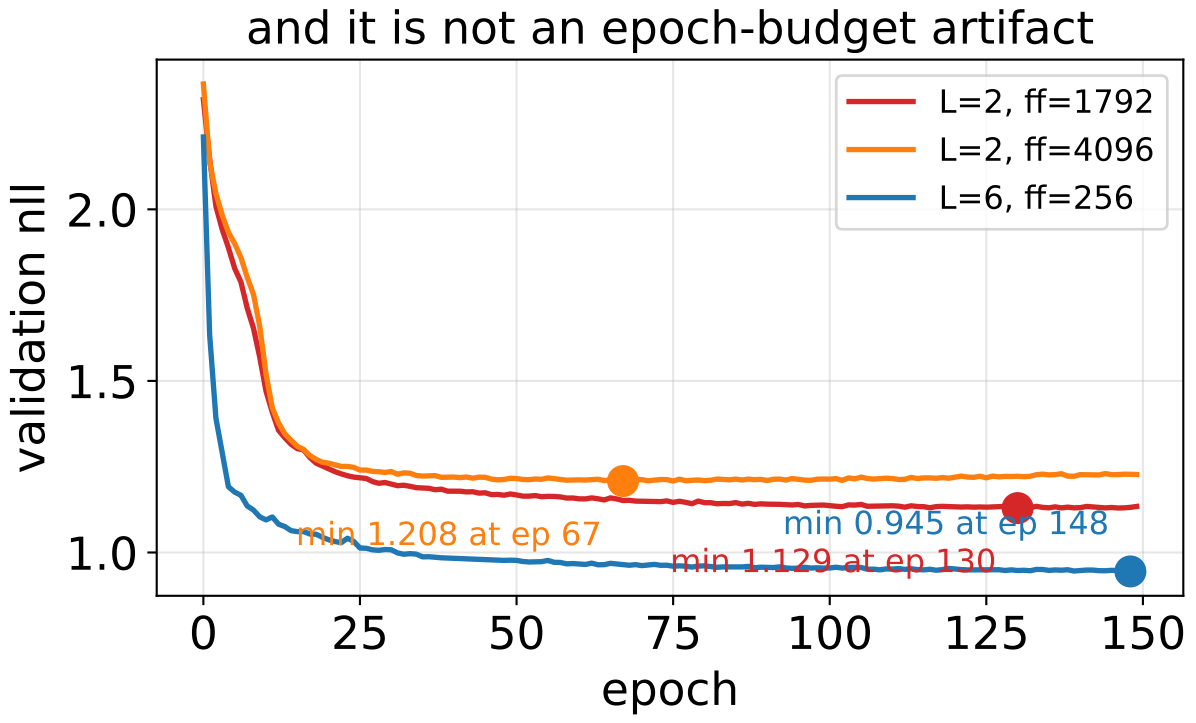
(d) HOLDS: the width axis reintroduces no depth-dependent L*



Finding 11 measured L*(T) at one MLP width, so this checks the flatness is not an artifact of ff=1024. It is not: both thin-deep cells clear 0.90 at EVERY depth bin at both densities (minimum 0.96, at T=5), and neither L=2 cell clears it at ANY bin. There is no T at which the ranking flips, so no depth-dependent threshold appears on this axis either. Each L=2 curve is also flat in T on its own terms -- ff=1792 runs 0.82-0.86 at rho=0.75 and 0.63-0.70 at rho=0.60, ff=4096 runs 0.68-0.70 and 0.34-0.41 -- so the deficit is a roughly constant offset per cell rather than something that grows with the number of rounds. That is the same signature finding 11 read as a fixed cost paid before chaining, now seen at ff=1792 and ff=4096 as well as at the ff=1024 it was measured on. Bin counts are 92-352 instances; these are the per-run depth bins, not the shared pool, so they are not the metric of record -- the margins here are far larger than that distinction.

Trade-off prediction (c) FAILS -- the one thing here that is unexplained



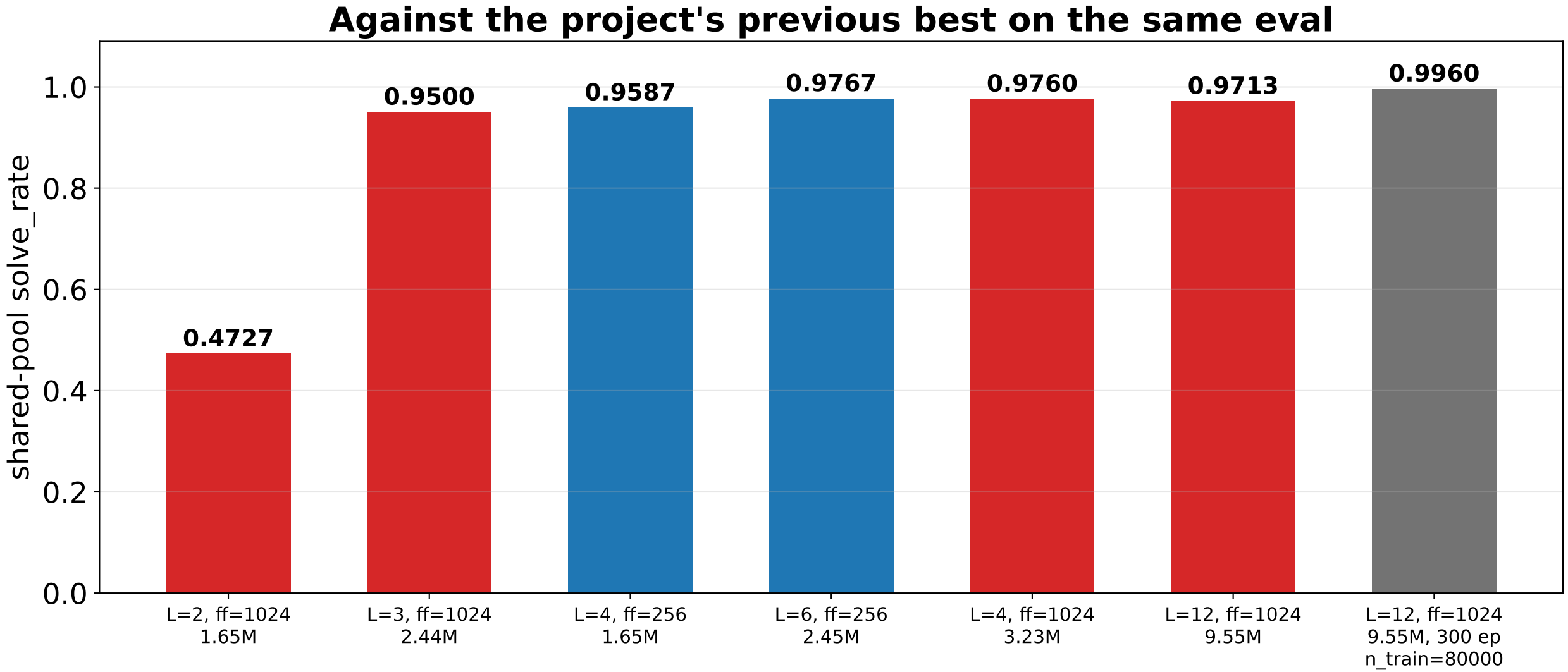


(c) pre-registered that solve_rate is monotone in ff at fixed L=2, as a check that the wide cells are trained long enough. It fails: ff=1792 scores 0.663 and ff=4096 falls back to 0.492 (-0.171, 112 to 369, p = 5e-33) -- a real drop, not noise, against a 0.02 tolerance. The sbatch said to read that as 150 epochs being too few for the wide cells and to stop. The right panel refutes that mechanism, and refutes it backwards: ff=4096's validation nll reaches its minimum of 1.208 at epoch 67, less than half the budget, and then drifts UP by 0.019 to 1.227 by epoch 150. A cell that was short of epochs would still be descending at the end; this one turned over at the halfway mark, which is mild overfitting rather than under-training. The thin-deep cell is the only one of the three whose minimum sits at the final epoch. So (c)'s failure is NOT confounded with the epoch budget, and the headline does not rest on it: (a) is measured entirely between L=2/ff=1024 and L=4/ff=256, neither of which is a wide cell. What remains unexplained is why width at L=2 should hurt at all. Reported, not resolved.

Scorecard

training (a)	**HOLDS**	L=4/ff=256 reaches 0.90 at matched params; measured 0.959 vs 0.473
training (b)	**HOLDS**	L=2/ff=4096 stays below 0.90 at 4.8M (1.97x the knee): 0.492
training (c)	**FAILS**	ff monotone at L=2: 0.473 / 0.663 / 0.492 -- drops 0.171 at ff=4096, p = 5e-33
training (d)	**HOLDS**	L*(T) flat on the ff axis: ff=256 clears 0.90 at every T (worst 0.96), L=2 at no T
pool (a)	**HOLDS**	depth beats width at 1.65M, paired p = 1e-203 (<< 0.001)
pool (b)	**HOLDS**	L=6/ff=256 >= L=3/ff=1024 at matched 2.44M: +0.0267, 65 to 25, p = 3e-05 (< 0.05)
pool (c)	**HOLDS**	L=6/ff=256 within 0.02 of L=12/ff=1024 at 3.9x fewer params: +0.0053, p = 0.37 (tied)

Seven pre-registrations across the two jobs, six held. The one failure, (c), is a sanity check rather than a hypothesis, and the training curves rule out the mechanism it was guarding against -- see the previous page. H1 (capacity) predicted that pool (a) would come in within 0.10 and that training (b) would clear 0.90; both went the other way, so the honest summary is that the L=3 knee is a serial requirement rather than a parameter count, with an unexplained width pathology at L=2 sitting next to it.



The grey bar is the standing best, 0.9960 at n=1500 (48912548 trained, 48999377 scored; 0.9955 when re-scored at n=6000), so it is on exactly the same pool and sample size as the coloured bars. It is NOT beaten here -- it has 4x the data and 2x the epochs, both of which finding 6 says are the strongest levers available. Nothing in this deck is a new state of the art, and it was not built to be one: every cell here is held at 150 epochs and n_train=20000 so that only the allocation moves. The comparison that matters is within the coloured bars, where L=6/ff=256 at 2.45M matches L=12/ff=1024 at 9.55M under an identical recipe. The obvious follow-up is to spend that efficiency rather than bank it: the best recipe so far has never been run thin-deep, and L=12/ff=256 would cost about 4.8M against the current 9.55M.